

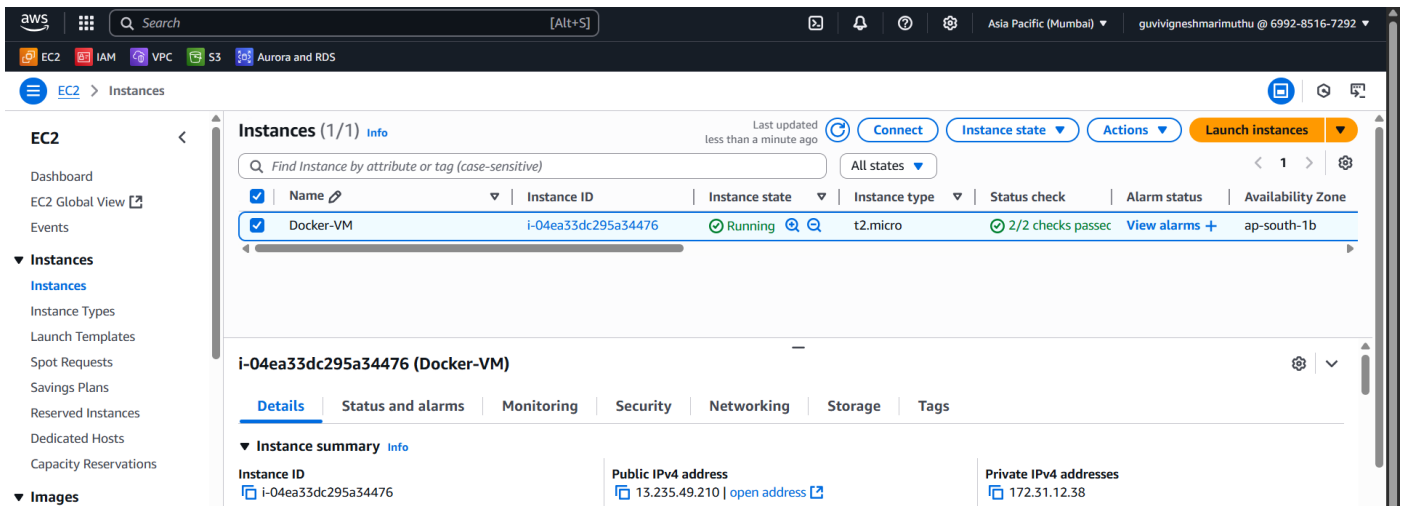
Docker Task -3

Task Description:

Create a custom docker image for nginx and deploy it using docker compose, where the volume bind mount should be at /var/opt/nginx location. Push the created custom docker image to your docker-hub.

Solution:

✓ 1. Launched my EC2 Ubuntu instance



✓ 2. Installed Docker and Docker Compose



✓ 3. Created my project folder



✓ 4. Created a custom HTML file for Nginx



✓ 5. Create a Dockerfile to build a custom Nginx image

```
root@ip-172-31-12-38:~/nginx-custom# nano Dockerfile
root@ip-172-31-12-38:~/nginx-custom# cat Dockerfile
# Base Image
FROM nginx:latest

# Remove default nginx HTML files
RUN rm -rf /usr/share/nginx/html/*

# Copy custom HTML to default nginx directory
COPY html/ /usr/share/nginx/html/
root@ip-172-31-12-38:~/nginx-custom#
```

✓ 6. Built the custom image using Docker

```
root@ip-172-31-12-38:~/nginx-custom# sudo docker build -t vigneshmarimuthu06/nginx-custom:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.584kB
Step 1/3 : FROM nginx:latest
latest: Pulling from library/nginx
dad67da3f26b: Pull complete
4eb3a9835b30: Pull complete
021db26e13de: Pull complete
397cc88dcd41: Pull complete
5f4a88bd8474: Pull complete
66467f827546: Pull complete
f05e87039331: Pull complete
Digest: sha256:dc53c8f25a10f9109ed5b59bda2d707a3bde0e45857ce9e1efaa32ff9cbc1
Status: Downloaded newer image for nginx:latest
----> 9a9a9fd723f1
Step 2/3 : RUN rm -rf /usr/share/nginx/html/*
----> Running in 07348c910463
----> Removed intermediate container 07348c910463
----> ae592053b0b0
Step 3/3 : COPY html/ /usr/share/nginx/html/
----> 0a9440e62f8f
Successfully built 0a9440e62f8f
Successfully tagged vigneshmarimuthu06/nginx-custom:v1
root@ip-172-31-12-38:~/nginx-custom#
```

i-04ea33dc295a34476 (Docker-VM)

PublicIPs: 13.235.49.210 PrivateIPs: 172.31.12.38

✓ 7. Created a docker-compose.yml to deploy the container

```
root@ip-172-31-12-38:~/nginx-custom# nano docker-compose.yml
root@ip-172-31-12-38:~/nginx-custom# cat docker-compose.yml
version: '3'
services:
  nginx:
    image: vigneshmarimuthu06/nginx-custom:v1
    ports:
      - "8080:80"
    volumes:
      - ./html:/var/opt/nginx
root@ip-172-31-12-38:~/nginx-custom#
```

✓ 8. Started the container using Docker Compose

```
root@ip-172-31-12-38:~/nginx-custom# docker-compose up --build -d
Creating network "nginx-custom_default" with the default driver
Creating nginx-custom_nginx_1 ... done
root@ip-172-31-12-38:~/nginx-custom#
```

i-04ea33dc295a34476 (Docker-VM)

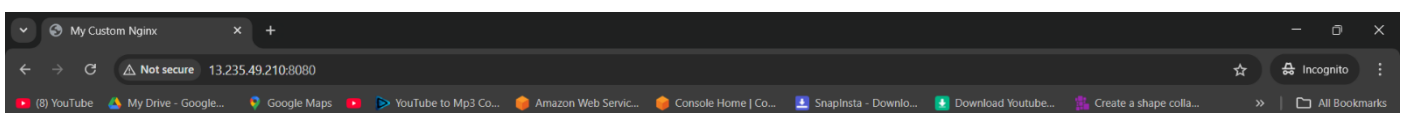
PublicIPs: 13.235.49.210 PrivateIPs: 172.31.12.38

✓ 9. Verified the container was running

```
root@ip-172-31-12-38:~/nginx-custom# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
c124bdd5c5c6	vigneshmarimuthu06/nginx-custom:v1	"/docker-entrypoint..."	18 seconds ago	Up 18 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	nginx-custom_nginx_1

✓ 10. Tested the website in browser



Hello, This is Vignesh's Custom Nginx Page!

✓ 11. Pushed the image to Docker Hub

```
root@ip-172-31-12-38:~/nginx-custom# docker login

USING WEB-BASED LOGIN
To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: QPLS-ZZVH
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credential-stores

Login Succeeded
root@ip-172-31-12-38:~/nginx-custom# docker push vigneshmarimuthu06/nginx-custom:v1
The push refers to repository [docker.io/vigneshmarimuthu06/nginx-custom]
403013d38d56: Pushed
257deee7ac4d: Pushed
cd38dca3d982: Mounted from library/nginx
d35594dd7e6d: Mounted from library/nginx
126eae18409: Mounted from library/nginx
6380429cac56: Mounted from library/nginx
13fcb2d303e8: Mounted from library/nginx
151f9fee563: Mounted from library/nginx
7fb72a7d1a8e: Mounted from library/nginx
v1: digest: sha256:d06acead770b6e3870f9193a7fb6dd44c316fbe31f14d89dc4d2a576981bed18 size: 2193
root@ip-172-31-12-38:~/nginx-custom#
```

i-04ea33dc295a34476 (Docker-VM)

PublicIPs: 13.235.49.210 PrivateIPs: 172.31.12.38

The screenshot shows the Docker Hub interface for the repository `vigneshmarimuthu06/nginx-custom`. The page includes a sidebar with navigation options like Repositories, Collaborations, Settings, and Billing. The main content area shows the repository details, including the tag `v1` which was pushed less than 1 minute ago. A table lists the tags with columns for Tag, OS, Type, Pulled, and Pushed. The `v1` tag is listed with OS `linux` and Type `Image`, pulled less than 1 day ago, and pushed 1 minute ago. There is also a section for Docker commands and a Build Cloud advertisement.

Tag	OS	Type	Pulled	Pushed
v1	linux	Image	less than 1 day	1 minute

- I successfully created a custom Nginx Docker image by replacing its default web content with a personalized HTML file.
- I built this image using a Dockerfile, then used Docker Compose to run the container.
- While deploying, I set a bind mount at `/var/opt/nginx` to meet the given condition.
- I tested the application using the EC2 instance's IP and port 8080.
- Finally, I pushed my custom image to Docker Hub at `vigneshmarimuthu06/nginx-custom:v1`.